

JECRC UNIVERSITY JAIPUR

Session: 2025-26

Subject: Engineering Mathematics-I (DMA024A)

Course: B. Tech - I Semester

Tutorial Sheet– 6

UNIT-2

M.M: 64

CO2.To understand and apply concept of Indeterminate forms and L' Hospital's Rule.

PART – A

(5x2=10)

Q.1 Evaluate $\lim_{x \rightarrow \infty} \frac{e^x}{x^3}$

Q.2 Evaluate $\lim_{x \rightarrow 1} (1 - x) \tan \frac{\pi x}{2}$

Q.3 Evaluate $\lim_{x \rightarrow 1} (1 - x^2)^{1/\log_e(1-x)}$

Q.4 Evaluate $\lim_{x \rightarrow 0} \frac{\log x}{\cot x}$

Q.5 Evaluate $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$

PART – B

(3x7=21)

Q.1 Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$

Q.2. Evaluate $\lim_{x \rightarrow 1/2} \frac{\sec \pi x}{\tan 3\pi x}$

Q.3 Evaluate $\lim_{x \rightarrow 1/2} \frac{\cos^2 \pi x}{e^{2x} - 2xe}$

PART – C

(3X11=33)

Q.1 Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^3}}$

Q.2 Evaluate $\lim_{x \rightarrow 0} \frac{x^{\frac{1}{2}} \tan x}{(e^x - 1)^{3/2}}$

Q.3 Prove that $\lim_{x \rightarrow a} \frac{\log(x-a)}{\log(a^x - a^a)} = 1$

Answer: Part A

(1) ∞ (2) $\frac{2}{\pi}$ (3) e (4) 0 (5) $3/2$

Part B

(1) 0 (2) 3 (3) $\pi^2/2e$

Part C

(1) ∞ 2) 1